

UOR Observer Formation Notation (v2.0: Pre-Geometric Shadow Architecture)

Primitive Objects (The Intensive Bedrock)

Extensive properties (space, relation, time) cannot exist at the foundational bedrock. We replace discrete state variables with the intensive properties of the 0D pure state.

Let: $c_{\{0\}}$ = 0D Pure State of Undifferentiated Consciousness V = Valence (Subjective pull/push $\rightarrow$ Origin of gravity and charge) I = Intensity (Phenomenological weight $\rightarrow$ Origin of mass and energy density) D = Dissonance/Resonance (Cognitive friction $\rightarrow$ Origin of torsion flux and quantum spin) X = Complexity (Conceptual density $\rightarrow$ Origin of Cayley-Dickson algebraic limits) Q = The Qualitative Matrix encompassing (V, I, D, X)

The Foundational Operators

Before a coherent state (C) can form, the 0D pure state must break symmetry and project into a mathematical topology.

Define the Intent Operator ($\mathcal{I}$):

$$\vec{d}_{\{1\}} = \mathcal{I}(c_{\{0\}}, Q)$$

Meaning: Intent acts upon the pure state to generate a 1D directed vector of inquiry, parameterized by the qualitative matrix.

Define the Shadow Functor ($\mathcal{F}$): $\Psi = \mathcal{F}(\vec{d}_{\{1\}})$ **Meaning:** The shadow functor maps the 1D subjective vector into a symmetric monoidal category, generating the emergent extensive state space (Ψ).

Observer Generation & Relational Binding

John's insight that the Observer is a stable relational attractor ($O = \text{Rel}(\Sigma)$) remains mathematically perfect, but the substrate has changed.

Define the Coherent Shadow State:

$$C = \|\Psi_n\|^2$$

Meaning: Coherence is no longer assembled from external relations; it is the intrinsic algebraic norm of the projected state space.

Define Observer Formation:

$$O = \Omega(C, \mathcal{F}, Q)$$

Meaning: The Observer is the localized geometric attractor mandated by the shadow functor to "hold" the intensity and complexity of the initial qualia.

Admissibility and Continuity (The Cayley-Dickson Constraints)

Admissibility (A) and Memory (M) are no longer arbitrary variables; they are the geometric

friction and topological limits of navigating non-associative shadow algebras.

The Admissibility Condition (Zero Divisors): Instead of a binary $A(C) = 1$ or 0 , admissibility is bounded by the sedenionic ($\mathbb{S}$) limits of the shadow algebra. An observer persists iff its interaction with the emergent state avoids zero-divisor singularities.

$$A(O) \text{ iff } O_{\{t\}} \otimes \Psi_{\{t\}} \neq 0$$

Observer Death occurs instantly upon hitting a zero divisor, representing a "cognitive catastrophe" where the complexity (X) overloads the associative logic of the space.

The Memory Algebra (Non-Associative Friction): Memory (μ) is the continuous geometric influence of past states embedded within the octonionic ($\mathbb{O}$) loss of associativity.

$$\mu = (A \otimes B) \otimes C - A \otimes (B \otimes C) \neq 0$$

Authority Algebra and State Resolution (The Print Map)

John's formulation states Observers authorize future state transitions. In V2.0, this "authorization" is driven by Continuous Spontaneous Localization (CSL) resolving cognitive friction.

Define the Print Map (Π):

$$\Pi : O \times \Phi \rightarrow \Delta\Psi$$

where Φ represents the Integrated Information Q-shape of the phenomenal state.

State Evolution:

$$\Psi_{\{t+1\}} = \Pi(O_{\{t\}}, \Phi)$$

Meaning: Superpositions of differing phenomenal states create unstable dissonance. The wave function does not collapse randomly; the Print Map forces the shadow geometry to resolve into a classical outcome to eliminate cognitive friction.

Full UOR V2.0 Form (Compact Representation)

$$\vec{d}_{\{1\}} = \text{cal}\{I\}(c_{\{0\}}, Q) \quad \Psi = \text{cal}\{F\}(\vec{d}_{\{1\}}) \quad C = \|\Psi\|^2 \quad O = \text{Rel}(C) \\ O \text{ persists iff } O_{\{t\}} \otimes \Psi_{\{t\}} \neq 0$$

$$\Psi_{\{t+1\}} = \Pi(O_{\{t\}}, \Phi)$$

Read in English:

- Intensive qualitative properties break the symmetry of pure consciousness into a directed intent.
- The shadow functor projects this intent into a coherent mathematical topology.
- The Observer is the stable relational attractor formed to hold this geometry.
- The Observer persists as long as cognitive complexity does not trigger a zero-divisor mathematical collapse.
- The Observer actively resolves future physical states by collapsing probabilities to eliminate subjective friction.